

Exercise (Week 7)

You are a data analyst at a university. You have been given student records and need to perform basic data management tasks.

You are required to perform task 1. Until 4.

1. Creating DataFrame

Create a data frame called `students` with the following information:

Name	Gender	Age	Test_Score	Attendance (%)
Aisha	Female	22	85	90
Bryan	Male	23	78	85
Chloe	Female	21	92	95
Daniel	Male	24	65	70
Emily	Female	22	88	88

R code to write :

```
# Create vectors for each column
```

```
# Combine into a data frame
```

```
# View the data frame
```

2. Access and Subset Data

Write R code to:

```
#Display the Test_Score column using both df['Test_Score'] and df$Test_score.
```

```
#Display the first 3 rows of the data frame
```

```
#Display students who are older than 22
```

```
#Display female students with Test_Score greater than 80
```

3. Add New Data

Name	Gender	Age	Test_Score	Attendance (%)
Lina	Male	20	95	92

Note : use rbind()

4.

```
students_na <- data.frame(  
  Name = c("Aisha", "Bryan", "Chloe", "Daniel", "Emily"),  
  Gender = c("Female", "Male", "Female", "Male", "Female"),  
  Age = c(22, NA, 21, 24, 22),  
  Test_Score = c(85, 78, NA, 65, 88),  
  Attendance = c(90, 85, 95, NA, 88)  
)
```

Write R code to:

- #Check for missing values (NA) in the data frame
- #Replace missing values in Age with the mean age
- #Replace missing values in Test_Score with the median test score
- #Replace missing values in Attendance with 0